

Cognitive Science — Module 02: Agents — Study Questions

1. What is the difference between a system and an agent? Can you give an example of a system that is NOT an agent?
2. What is a generative model? Explain in your own words how it represents the agent's "story about the world."
3. What are beliefs and preferences in Active Inference? Are these conscious? Does a bacterium "believe" things?
4. Describe the predict-compare-update cycle. Create your own novel example showing all four steps (predict, compare, compute error, minimize error).
5. What are the two ways to minimize prediction error? Which one changes the world and which one changes the model?
6. Why does the body shiver when it's cold? Explain this using the language of predictions, prediction errors, and active inference.
7. What is a self-model? Why does your brain need to model itself in addition to the external world?
8. What happens when the self-model breaks down? Describe one clinical example (alien hand syndrome, depersonalization, or phantom limbs) and explain it as a self-model failure.
9. Where does agency begin on the spectrum from atoms to humans? Is a virus an agent? Is a plant? Defend your answer.
10. What makes human agency different from bacterial agency? What specific property of human generative models enables reflective thought?
11. How does the "morning routine" example demonstrate that the generative model is always running — even before you open your eyes?
12. Compare a thermostat, a bacterium, and a human student as agents. What does each one predict, and what actions does each take to minimize prediction error?
13. What is meta-inference? Why might the ability to doubt your own beliefs be the most powerful feature of human cognition?
14. If an AI chatbot can predict user messages and generate appropriate responses, does it qualify as an agent under the Active Inference definition? Why or why not?

15. Critically evaluate: If agency is defined by having a generative model, does this definition include too many things (making it meaningless) or is it appropriately broad? What are the strengths and weaknesses of this definition?
16. How does the concept of "niche construction" — organisms actively modifying their environment — relate to Active Inference? Give an example of a beaver building a dam and explain it as active inference extending beyond the body boundary.
17. What is the relationship between embodiment and agency? Why can't a "brain in a vat" be a full agent under Active Inference? What specific role does the body play in completing the perception-action cycle?
18. How does Active Inference explain the "rubber hand illusion"? When a fake hand is stroked in synchrony with your real (hidden) hand, your self-model temporarily incorporates the fake hand. What predictions and prediction errors drive this remarkable phenomenon?
19. Consider a group of people coordinating without a leader — for example, an audience that spontaneously begins clapping in unison. Can the group be considered an agent? What would its generative model look like, and where is its Markov Blanket?
20. How might an Active Inference agent develop "habits" — behaviors that become automatic and require less computational effort over time? What changes in the generative model when a behavior becomes habitual, and why is this computationally efficient?